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Title (required)

Estimating dynamic functional connectivity using Hidden Markov Model with low-rank approximation

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Introduction (required)

Hidden Markov Models (HMMs) are widely used to characterize dynamic Functional Connectivity (dFC) in both resting-state and task fMRI.^{^1} Due to the high dimensionality of regions of interest, regularization approaches are commonly used to avoid overfitting. For example, Principal Component Analysis (PCA) projects the data to lower dimensions,^{^2} and Graphical Lasso (GL) imposes sparse constraints on the inverse covariance.^{^1} However, both approaches have limitations: PCA is effective for reducing dimensionality, but the covariance in the original space is difficult to recover. For GL, the regularization can lead to state imbalance, where states that appear less frequently may produce unstable predictions for unseen data. Moreover, because sparsity is imposed on the inverse covariance matrix under a positive-definiteness constraint, the GL objective is most naturally enforced within Expectation–Maximization (EM)–based estimation, where the M-step admits an exact and convex solution. In contrast, incorporating such constraints into gradient-based discriminative training generally breaks the original GL formulation and complicates optimization.

In this abstract, we present an alternative way to regularize fMRI data using a mixture of factor analyzers (MFA).^{^3} The starting point for MFA is to define a state-dependent projection that maps the original data into lower dimensions. As a result, the covariance matrix can be written as a combination of a low-rank term W and a positive diagonal term Ψ . We find that this method has several advantages. First, compared with PCA, the covariance matrix in the original space can be reconstructed. Second, compared with GL, MFA generates more interpretable states with fewer parameters, which naturally mitigates overfitting. Additionally, the states are more balanced, and the differences between states are enhanced. Numerically, we evaluate our results using Kullback–Leibler (KL) divergence and the Bayesian Information Criterion (BIC).

Methods (required)

The fMRI data were acquired on a 7T Siemens MAGNETOM Terra MR scanner.^{^4} The dataset contains 46 cognitively normal subjects and 41 subjects with mild cognitive impairment (MCI). Each subject underwent 1–3 task fMRI scans on pattern separation, resulting in a total of 235 scans. The functional MRI runs were collected using a multi-band gradient-echo EPI sequence with TR = 1530 ms and 510 time points. All subjects underwent preprocessing, including slice-time correction, realignment, and normalization to MNI space. The fMRI data were denoised by regressing out CSF and gray-matter signals and then parcellated into 32 regions using the CONN-32 atlas (<https://web.conn-toolbox.org/>). For each region, the meantime series was extracted, yielding a 32-dimensional representation. Finally, the data have size $235 \times 510 \times 32$.

We build an HMM with number of hidden states $K = 3$ using four different models. The first model is a commonly used HMM that takes all features $p = 32$ as input. The second model also uses all features but applies GL regularization with $\lambda = 0.01$. The third model uses PCA to reduce the feature dimension to $q = 12$ and then builds HMM on the reduced-dimensional data. The fourth model uses MFA to reduce the dimension to $q = 4$. The main difference between the third and fourth models is that PCA applies the same dimensionality-reduction technique to all three hidden states, whereas MFA uses different reductions for each state. In practice, all data from the 235 scans are used, and the unknown parameters are estimated using the EM algorithm. To avoid local minima, we perform 100 random initializations for each method, run EM until convergence, and select the solution that yields the maximum likelihood.

Figure 1 lists the four models considered in this paper. Figure 1(a) shows the covariance matrices, state occupational probabilities, and unknown parameters for the traditional HMM. 32 networks from eight categories are chosen: Default Mode (DMN), Sensorimotor (SM), Visual (VIS), Salience (SN), Dorsal Attention (DAN), Frontoparietal (FPN), Language (LAN), and Cerebellar (CB). Number of unknown parameters does not include the transition matrix, and symmetry is considered for the covariance matrix. Figure 1(b) presents the corresponding outputs for the GL model with $\lambda = 0.01$. While the covariance matrices appear similar, we observe that the occupational probabilities become unbalanced, as the probability for the third state is reduced. From a parameter perspective, GL does not substantially reduce the number of parameters but already introduces a significant imbalance problem. Figure 1(c) shows the results for PCA with reduced feature dimension $q=12$. Using this method, the first two states become almost identical, indicating a loss of information in the original dimensions. Finally, Figure 1(d) shows the MFA model with $q=4$, where the number of parameters is reduced while interpretability is maintained. Since PCA does not generate an interpretable covariance matrix in the original space, we only compare GL and MFA in the later analysis.

Methods Figure (Optional)

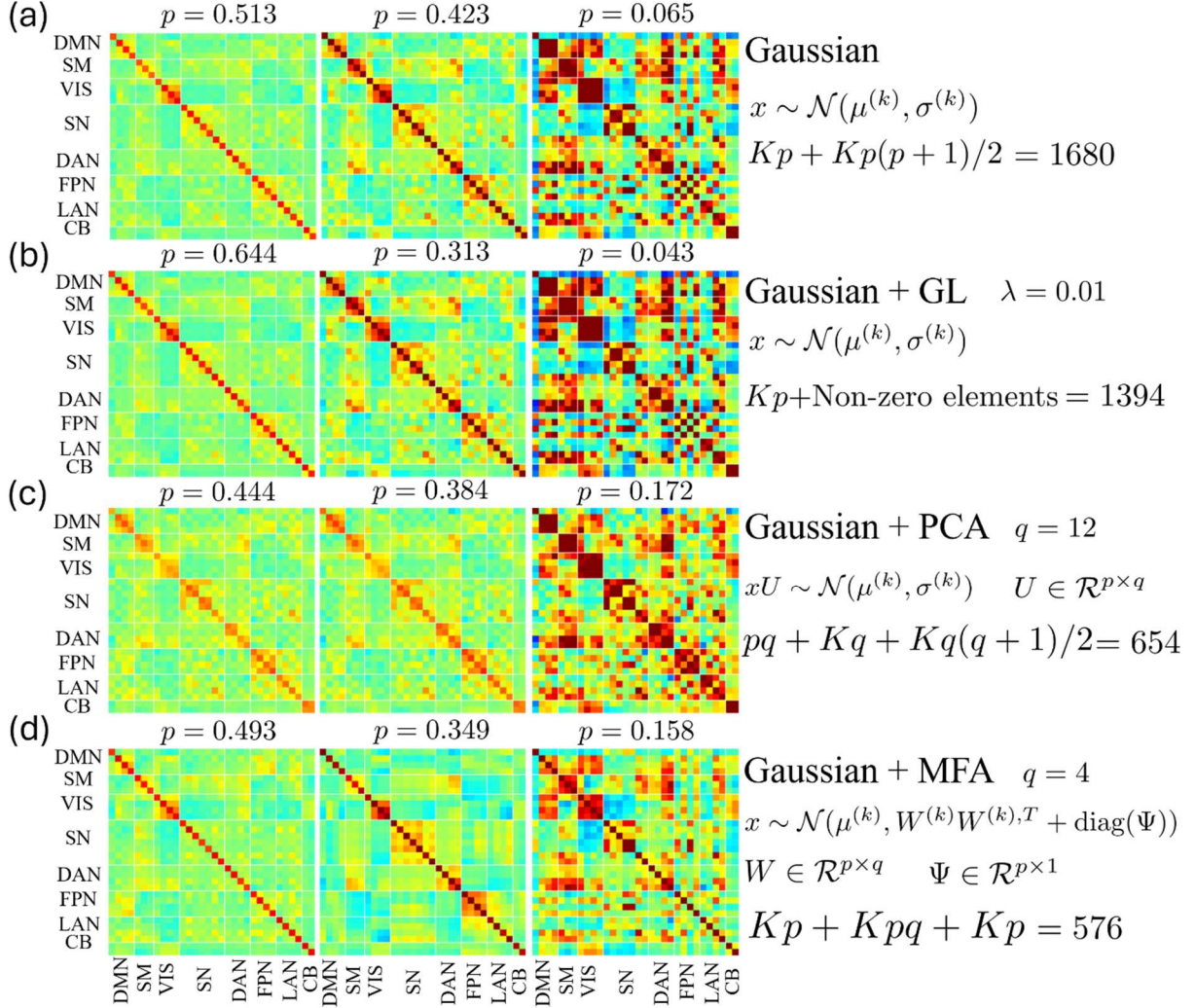


Figure 1: (a) dFC for the Gaussian HMM on fMRI data with 32 networks. The occupational probability is shown on the top, while the number of parameters is shown on the right. (b) Gaussian HMM with GL regularization with $\lambda = 0.01$. (c) PCA with $q=12$. (d) MFA with $q=4$. MFA is the most effective at reducing the number of parameters, maintaining similar dFC, and producing more balanced states.

Results (required)

To further demonstrate the advantages of MFA, we first compute the differences between the states. In HMMs, the states are commonly determined by unsupervised methods such as EM. A desirable property is that the states should be as distinct as possible, which typically serves two purposes. First, if two hidden states are nearly identical, this may indicate that the chosen number of hidden states is too large. Second, when HMMs are implemented for different groups, such as patients at different disease stages, the ability to match the K hidden states across groups becomes important, and this task is easier when the states are more distinct. We also note that even if one state differs substantially from

the others, it contributes little to the overall difference if it rarely appears. Therefore, we define the weighted difference as $\sum_{ij} p_i p_j D_{ij}$, where p_i measures the occupational probability for state i and D_{ij} measures the difference between states i and j . In practice, we use the KL divergence to compute D_{ij} . This quantity measures the expected difference between two randomly selected time points.

Figure 2(a) shows the weighted KL divergence versus the number of hidden states. We find that the HMM without regularization and the HMM with GL $\lambda = 0.01$ do not produce sufficient differences between states for the same value of K , whereas MFA generates substantially larger differences, indicating the advantages of MFA for this problem.

Second, we examine the BIC on a small dataset. The definition for BIC is shown in Figure 2(b). The general goal is to obtain a BIC value that is as small as possible. Large models fit the data well and thus yield a small first term, but they also contain a large number of parameters, causing the second term to act as a strong penalty. In practice, because the first term is linearly proportional to the data size while the second term is logarithmically proportional to the data size, large datasets tend to favor large models. This pattern is commonly observed when BIC is computed on the full dataset, often resulting in the selection of large models.⁵ However, for prediction problems that involve splitting data into training, evaluation, and testing sets, we want the model to perform reasonably well on small datasets. Therefore, we evaluate the model using data with a size comparable to the testing set. Specifically, among the 87 subjects, we select 10 subjects to train the model and another 10 subjects to compute the probabilities. This process is repeated 100 times to avoid statistical variation.

The BIC results are shown in Figure 2(b). MFA yields a much smaller BIC value compared with the traditional HMM and the HMM with GL. To better illustrate MFA performance, Figure 2(c) focuses exclusively on the MFA results. We note that the minimum BIC value occurs at $q = 4$ and $K = 4$.

Results Figure (Optional)

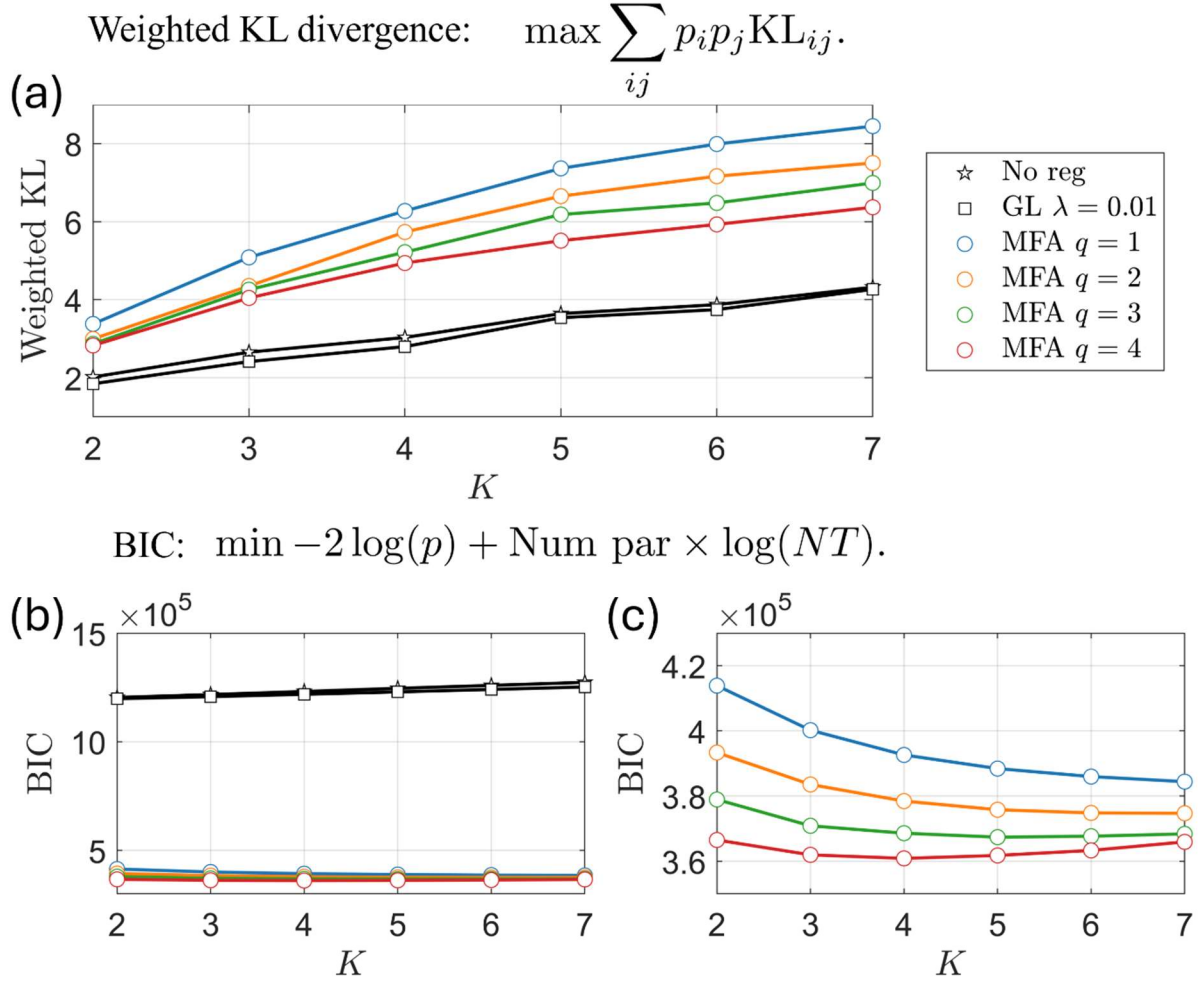


Figure 2: (a) Weighted KL differences evaluated using all data. Larger differences indicate better hidden-state separation. MFA benefits more from separated states compared with other methods. (b) BIC computed across different methods using data from 10 subjects. MFA shows a much smaller BIC due to its reduced number of parameters. (c) Same as (b) but focused on the small-BIC region. The smallest value occurs at $K=4$ and $q=4$.

Conclusion (required)

In conclusion, we introduce a regularization framework based on MFA for estimating dynamic functional connectivity. This approach substantially reduces the number of parameters relative to traditional methods, helping to mitigate overfitting while preserving the structure of the connectivity matrices. A key practical advantage is that the MFA formulation is naturally compatible with gradient-based optimization: as for any low-rank matrix W and any positive diagonal term Ψ , the resulting covariance matrix remains positive definite. In addition, MFA offers interpretability from a different perspective by identifying the subset of network features that predominantly drive state formation—a useful property given the large number of networks in fMRI studies. At the same time, MFA

sacrifices interpretability regarding direct versus indirect connections, which is a strength of GL.

References/Citations (required)

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